■ ■ EASY START (you can begin THIS WEEK)

1. DNA Sequence Analysis (Bioinformatics)

What to do:

- Take a gene sequence (from NCBI database)
- Analyze:
- GC content
- Mutation differences
- Use Python (Biopython)

Tools:

- Python
- Biopython

Output:

- Simple report + graphs

Why it helps:

- Bioinformatics = high demand in Ireland

2. Disease vs Gene Study

Example:

- Compare gene mutations in:
- Normal vs cancer cells

Tools:

- NCBI / UniProt databases
- Excel / Python

Output:

- Case-study style project

■ ■ INTERMEDIATE PROJECTS (2nd year level)

3. Drug-Target Interaction (Simulation)

What to do:

- Pick a disease protein (like COVID spike or cancer protein)

- Simulate how a drug binds to it

Tools:

- AutoDock

- PyMOL

Output:

- Binding affinity results + visuals

This is VERY impressive for biotech

4. AI-Based Disease Prediction

Example:

- Predict diabetes or heart disease

Steps:

- Use dataset (Kaggle)

- Train model (Python)

Tools:

- Python (pandas, sklearn)

Output:

- ML model + accuracy report

Combines biotech + AI → high value

5. Biomedical Device Concept

Example ideas:

- Smart glucose monitor

- Low-cost heart rate tracker

Even a prototype or simulation works

Bonus:

- Use Arduino (if possible)

■ ■ ADVANCED PROJECTS (3rd year level)

6. CRISPR Gene Editing Simulation

What to do:

- Simulate gene editing (not lab, software-based)

Tools:

- Benchling
- SnapGene

Output:

- Edited gene sequence + explanation

Very strong for genetics

7. Cancer Genomics Analysis

What to do:

- Analyze real cancer datasets
- Identify mutation patterns

Tools:

- Python / R

This is research-level work

8. Medical Image Analysis (AI)

Example:

- Detect tumors from MRI/CT images

Tools:

- Python + OpenCV / TensorFlow

Output:

- Model that detects abnormalities

VERY high demand in Ireland

■ ■ BEST COMBO (if you want top universities)

Pick ONE from each category:

- Bioinformatics → DNA analysis
- AI → Disease prediction
- Research → Drug docking / genomics

■ How to present your project (IMPORTANT)

Don’t just “do it”—document it:

- Make:
- GitHub repository
- Report (PDF)
- Simple presentation

This is what universities actually evaluate